



Reduction of Bubble-Induced Defect in Semiconductor Lithography Process

Kyoung-Whan Oh¹, Takashi Sasa¹, Seok Heo¹, Daejung Kim¹, Ouiserg Kim¹,
and Jung-Hyeon Kim²✉

¹ Samsung Electronics, 1-1 Samsungjeonja-ro, Gyeonggi-do 18448, Republic of Korea
kw208.oh@samsung.com

² Kumoh University, 61 Daehak-ro, Gyeongsangbuk-do 39177, Gumi-si, Republic of Korea
trizkim@kumoh.ac.kr

Abstract. In this study, methods for suppressing the occurrence of air bubbles in the photoresist (PR) dispensing apparatus of photo lithography system in semiconductor manufacturing process were analyzed. The proposed ways are intended to reduce product defects by targeting the specific areas where air bubbles are generated in the manufacturing equipment. The semiconductor industry faces the major challenge of reducing product defects by suppressing nano-particle contamination and bubble generation in PR. The width of the circuit pattern has narrowed as exposure technology in scanner equipment has improved especially using extreme ultraviolet (EUV) light sources. As a result, managing the smaller size of particles and bubbles has become increasingly difficult. Currently, semiconductor manufacturing equipment can manage wet particles and bubbles as small as 19 nm, and the industry continues to push for even smaller sizes. This paper aims to analyze the root cause of bubble generation in the dispensing system that delivers PR from the bottle to the nozzle for coating the wafer. By employing TRIZ inventive principles, the analysis leads to a detailed and efficient improvement plan that can be applied to existing mass production facilities to achieve zero instances of air bubbles. Furthermore, the proposed device designed to eliminate the root cause of air bubbles is incorporated into the newly developed facility, effectively reducing air bubbles and particles and ensuring high cleanliness.

Keywords: TRIZ · Lithography · Photoresist · Bubble · Defect

1 Introduction

Recent semiconductor photo lithography processes have been able to drastically shorten the process sequence and produce more delicate patterns by applying extreme ultraviolet (EUV) exposure equipment. This has made it possible to produce products that can integrate more semiconductors per wafer. Therefore, nano-particle contamination remains one of the major challenges in the semiconductor industry [1]. The critical particle size that the semiconductor industry has managed so far is in the range of 26–32 nm. However, as the current EUV photo process gradually reduces the size of the structural pattern, the

industry must be able to manage particles as small as 19 nm. Furthermore, this critical particle size is continuing to get smaller and smaller. This presents a technical limit that is difficult to achieve with existing commercial systems currently in use today.

The photoresist (PR) is a key chemical solution used in the process of drawing semiconductor circuits on semiconductor wafers through exposure. When PR is applied on a wafer and exposed to light engraved with a specific pattern, PR undergoes a chemical change, resulting in the formation of a circuit. Since particles placed on the pattern become product defect [2] sources, improving particle contamination in the manufacturing process is a main factor in improving process yield [3]. Previous studies have attempted to control the thickness of the layer by using a small amount of PR to reduce the amount of expensive PR used [4, 5].

In this study, we aim to improve the PR dispensing system to reduce product defects and improve productivity. Specifically, we analyze the root cause of air bubbles in fluids, which have long been a problem in the lithography process for manufacturing, and seek ways to improve the system to eliminate them. The study presents efficient improvement plans for existing mass production equipment and proposes new device for newly developed equipment to minimize the generation of bubbles.

2 Problem Analysis

The main focus of research and system design for particle reduction in the PR dispensing systems has been to improve PR stagnation. PR is composed of high molecular weight compounds in addition to solvents, and when the chemical solution is stagnant in the tube or parts, the high molecular weight compounds in PR combine with each other to form gel-like particles. Bubbles or particles can act as seeds for the formation of these polymer particles. When these particles are ejected onto the wafer, they can create defects in the product [6].

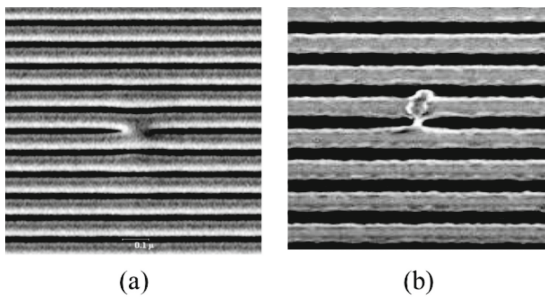


Fig. 1. (a) Line open defect due to a bubble and (b) a nano-bridge defect because of a particle associated bubble

In this section, we will employ TRIZ to comprehend the mechanism of air bubble formation during the movement of chemical solutions through facility tubing. Our goal is to analyze the root cause of bubble generation and devise a solution that can effectively

minimize the defects caused by bubbles as illustrated in Fig. 1(a) and the defects triggered by particles that adhere to the zeta potential of bubbles as shown in Fig. 1(b).

2.1 Root Cause Analysis

One of the recurring issues during PR transportation process is the formation of bubbles in the pump inlet region. These bubbles can flow into the pump, causing equipment stoppages and resulting in product defects when they are discharged onto the wafer. To understand the root cause of this problem, we conducted a Root Cause Analysis (RCA) starting from the observed bubbles and identified the causal relationship leading to the formation of nano bubbles, as illustrated in Fig. 2.

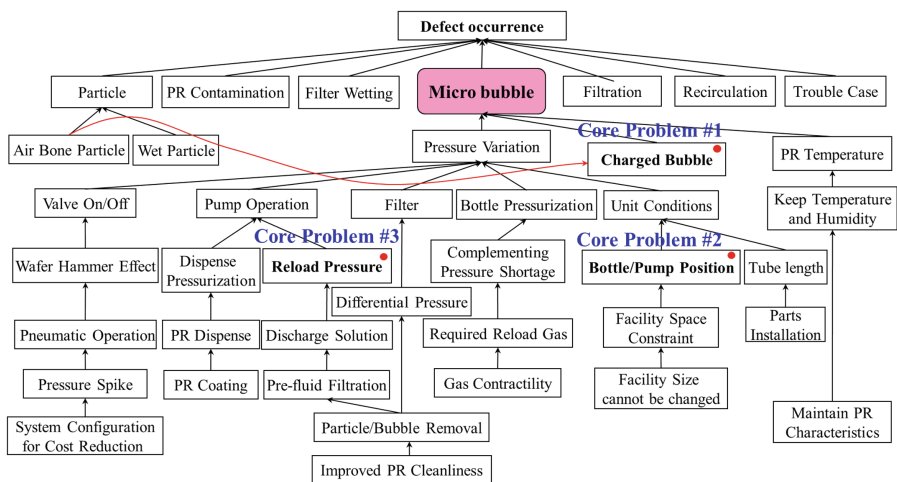


Fig. 2. The root cause analysis and three core problems for nano bubble generation.

Our analysis revealed three core problems associated with nano bubbles. The first one is the presence of charged bubbles. It was confirmed that negatively charged bubbles could cause particle defects by attracting positively charged particles. We decided that further understanding and analysis were needed for the second core problem, which is the correlation between the position of the bottle and the pump, as well as the third core problem, which is the differential pressure generated during PR reload operation of pump. We planned to verify these through experiments.

2.2 Experimental Verification

To further analyze core problem 2 and 3 and identify the causes of bubbles, we constructed a test device shown in Fig. 3. Through this, after reviewing the bubble generation-related factors by RCA, we intend to quantitatively review the location of the PR bottle and the pressure variation of the fluid, which are judged to be most relevant. To analyze

the correlation between the pump and the bottle as install location point of view, fluid discharge experiments were conducted under different height conditions.

As shown in Fig. 4, experiments confirmed that the average pressure of the chemical solution measured based on the pump inlet was directly proportional to the installation height of the bottle. The gauge pressures were measured at the pump inlet and recorded as + 4.8 kPa, -1.3 kPa, and -8.1 kPa for each bottle position (+0.5 m, -0.2 m, and -0.7 m height, respectively). By applying the simplified form of the Bernoulli equation, represented by Eq. (1) with P - pressure, ρ - density, g - gravity acceleration, v - velocity, the calculated results were found to be very close to the measured values, which were + 4.9 kPa, -1.9 kPa, and -6.9 kPa. Therefore, it can be concluded that the observed pressure difference is due to the gravitational potential energy.

$$P_1 + \frac{\rho v_1^2}{2} + \rho g z_1 = P_2 + \frac{\rho v_2^2}{2} + \rho g z_2 \tag{1}$$

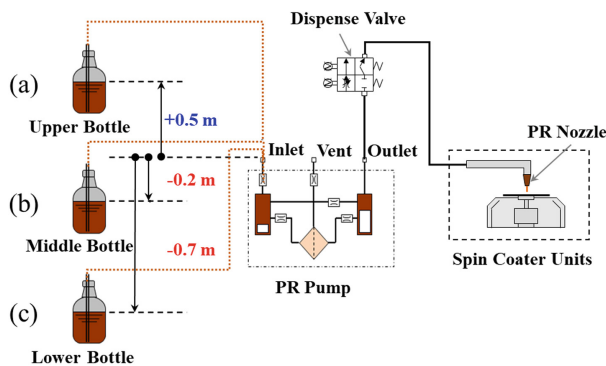


Fig. 3. Schematic of the PR dispensing system and an illustration of test conditions at different bottle positions (a), (b) and (c).

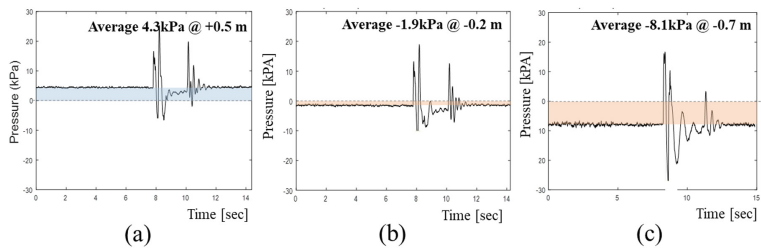


Fig. 4. Pressure variation during a dispensing cycle at the pump inlet for different bottle positions.

Furthermore, in order to investigate how the difference in physical pressure of the chemical solution, depending on the physical mounting position of the part, affects bubble generation, a test was conducted by dispensing the chemical solution 1000 times in each bottle position condition. Through this experiment, it was observed that when

the average pressure of the solution measured at the pump inlet was 4.8 kPa, -1.3 kPa, and -8.1 kPa, respectively, the amount of bubbles generated was 0.00 mL, 0.03 mL, and 0.07 mL, respectively. This bubble trend was found to be inversely proportional to the pressure as shown in Table 1. This confirms that the solubility of a gas in PR can be explained using Henry's law, which states that the solubility of a gas in a liquid is proportional to the partial pressure of the gas in equilibrium with the solvent.

Table 1. Comparison of test results regarding pump inlet pressure and generated bubble volume under different height conditions during a dispensing cycle.

Bottle Height (m)	Theoretical Pressure (kPa)	Average Pressure (kPa)	Minimum Pressure (kPa)	Bubble Volume (mL)
(a) Upper: + 0.5	4.9	4.8	-8.4	0.00
(b) Middle: -0.2	-1.9	-1.3	-11.5	0.03
(c) Lower: -0.7	-6.9	-8.1	-18.0	0.07

Based on the experimental results, we can confirm that maintaining a positive pressure in the chemical solution inside the tube, including the pump, is the first key design parameter to effectively reduce bubble generation.

2.3 Improvement of Existing System

In this section, we will review the improvement plans for existing mass production facilities based on the experimental results from the previous section. Since there are limited elements that can be changed in existing facilities, we have identified modification items to suppress bubble generation while minimizing potential changes, using the inventive principles of TRIZ. The identified items are summarized in Table 2.

The first modification item is to change the position of PR bottles upward. Due to safety reasons, PR is supplied in one-gallon glass bottles, which are stored in a cabinet divided into upper and lower parts. There are a total of 16 bottles stored in the cabinet. To increase the pressure of the fluid by utilizing potential energy, all the PR bottles are placed on top of the cabinet.

Table 2. TRIZ invention principles and modifications to improve fluid pressure.

Principle No	Inventive Principle	Corrective Action
#12	Equipotentiality	Change bottle installation height upward
#5	Merging	Reduce bottle quantity (16 → 8ea)
#11	Prior Compensation	Increase buffer tank size
#34	Rejecting, Discarding	Eliminate unnecessary valves and unions

Second, due to space constraints at the top of the cabinet, the number of PR bottles is reduced by half from 16 to 8. In supplying one type of PR, the two-bottle system, in which two bottles are alternately used, was applied in the existing system as shown in Fig. 5 (a). The facility can continue the process of supplying PR by replacing the supply of PR with a second bottle when PR supply in the first bottle is exhausted. Besides of this, the facility secures time to replace the exhausted PR bottle. In spite of these essential functions, in order to overcome space limitations, the two-bottle system is changed to a one-bottle system as shown in Fig. 5(b).

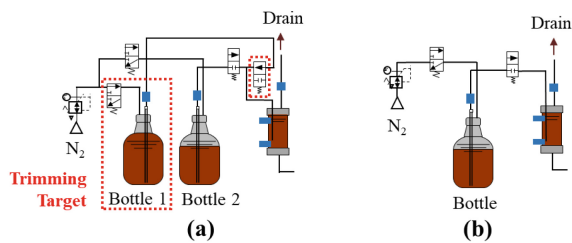


Fig. 5. Optimization of PR bottle system from (a) two-bottle system to (b) a one-bottle system.

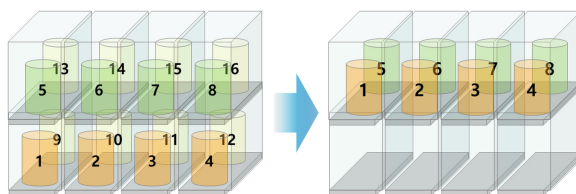


Fig. 6. Schematic illustration of PR bottles before (left) and after (right) optimization.

By combining the first and second changes mentioned above, we can modify the system design to overcome space constraints and retrofit the installation of PR suppliers on top of the cabinet to increase the potential energy of PR as shown in Fig. 6.

Third, the capacity of the reservoir tank for storing PR as a buffer is increased. Therefore, in order to secure adequate time for bottle replacement when PR is exhausted in one-bottle system, the capacity of the reservoir tank is greatly expanded from 50 mL to 200 mL as shown in Fig. 7.

In conclusion, the last modification to the existing system is to remove redundant parts such as valves, unions, and tubes that are no longer necessary due to the simplification of the system. These parts can cause additional pressure drop and particle generation, so their removal can improve the flow path. By implementing all four modifications, the fluid pressure can be increased by 8 kPa using physical potential energy, which is expected to reduce the occurrence of bubbles and improve the operation rate of the facility while minimizing defects caused by bubbles. The results of the facility improvements are presented in Chapter 3.



Fig. 7. The reservoir tank optimization of numbers and size before and after tanks' schematic illustration.

2.4 Modified Modeling of Smart Little People

One of the fundamental requirements in developing a new photo track facility is to install a considerable number of PR bottles in a confined space, without any limitations on the installation height, to provide various types of PR. To comprehend the pressure drop occurring in a 7–10 m long tube from the bottle to the pump, a small human model is utilized to analyze the system.

The modified version of the smart little people (SLP) model can provide a simplified and intuitive way to understand the pressure drop in the system as compare to classical SLP method [7]. By using a ringer as the reservoir tank, a medical tube as the long PR tube, and two syringes as the PR pump, it becomes easier to visualize the flow of PR and the pressure drop that occurs. Using the model of SLP as PR allows for a better understanding of how the fluid moves through the system and the effects of gravity on the pressure drop. This modified version can be a useful tool in the design and optimization of new photo track facilities.

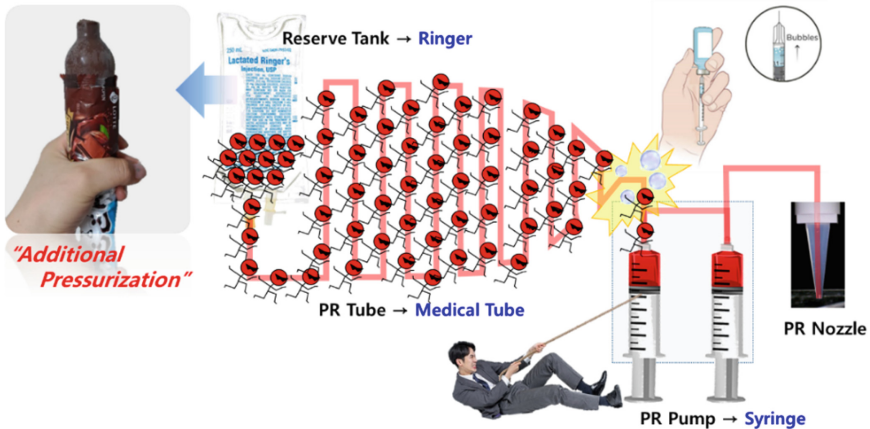


Fig. 8. The modeling of Smart Little People with alternative system for PR dispense units.

From this observation, it can be inferred that bubbles might occur at the syringe inlet, similar to the bubbles generated when filling a syringe with liquid from an ampoule, as shown in Fig. 8. To prevent the occurrence of such bubbles in the system, it is expected

that additional pressurization can be performed, as if the ring is squeezed by hand, like a pouch-type ice cream, to prevent pressure drop of the fluid.

This solution is expected to prevent the occurrence of bubbles in the system and improve the flow of fluid by compensating for pressure drops inside the tube. The system design of the auxiliary pressurization device is shown in Fig. 9 and it is focused on controlling the pressure inside the tube instead of the reservoir tank. The effectiveness of this solution will be evaluated further through experimental tests in the next chapter.

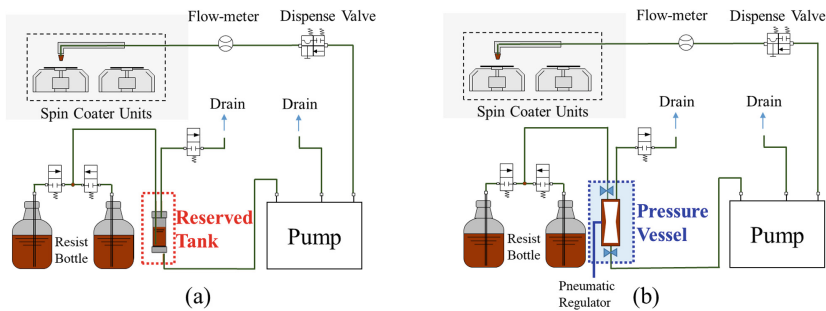


Fig. 9. (a) Conventional PR dispense system and (b) new system with the novel pressure vessel.

The introduced pressure vessel in this study consists of an inner tube with a four-leaf clover-shaped cross section and a circular outer tube for injecting air to control it in a dual structure coaxially. The inner tube is used for the flow and conveyance of PR while the outer wall of the outer tube has a tube through which gas can be inserted and discharged to supply and remove gas with a desired pressure profile. The gas is injected between the inner and outer tubes as shown in Fig. 10 which causes the inner tube to contract and expand, indirectly controlling the fluid pressure of PR in the inner tube.

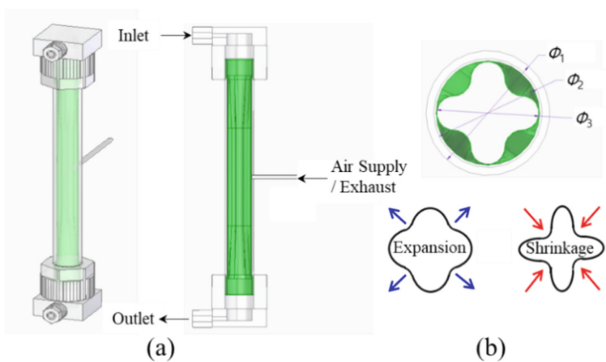


Fig. 10. (a) Pressure vessel configuration (b) Cross-section of pressure vessel as expansion and shrinkage operation.

In particular, the auxiliary pressurization device is designed to maintain positive pressure during PR reload operation, with effective control performance ensured by

feedback from the pressure sensor built into the pump. The contraction and expansion of the inner tube can be effectively controlled through feedback control of the pressure sensor and valve operation at the container inlet and outlet. The inner tube is optimized and designed for durability and control performance during operation.

3 Evaluation Results

This section presents the tools used to analyze the root cause of the bubble generation problem and the proposed solutions for improving air bubble problems in mass production lines. The experiments conducted were aimed at identifying the underlying factors contributing to the problem, and the results obtained provided a basis for designing effective solutions. The proposed solutions include remodeling of existing equipment and the application of newly designed parts to new facilities. The effectiveness of the proposed solutions was confirmed through experiments conducted in actual mass production lines. Overall, this section highlights the significance of a systematic approach to identifying and addressing problems in semiconductor manufacturing processes.

3.1 Results of Application

Air bubbles generated at the inlet of the PR pump are a major issue in the photolithography process for semiconductor manufacturing, as they frequently trigger equipment alarms and cause unexpected shutdowns. Chapter 2.2 of this study confirmed that these bubbles are produced when the pressure of the chemical solution drops to negative during the PR reload operation, which refills the stage inside the pump after PR dispensing operation.

In order to address this issue, the modification items proposed in Chapter 2.3 were implemented in two photolithography facilities, and the occurrences of bubble alarms were compared and reviewed for one month before and after the improvements. The results showed in Fig. 11 that the alarm trend was completely eliminated. Additionally, long-term monitoring of the same facilities for an additional six months confirmed that they continued to operate without any bubble alarms.

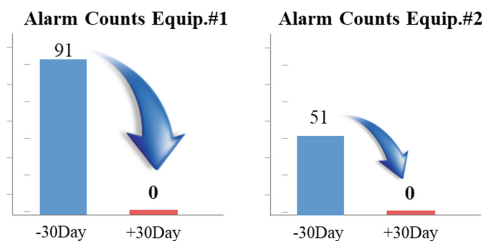


Fig. 11. Comparison of the number of bubble induced alarms for a month with 4 photo track equipment before and after improvement.

3.2 Results of Experimentation

Reducing the number of PR bottles in a facility and placing them all in an upward position in the latest photo lithography process, which requires a variety of semiconductor manufacturing, is realistically very difficult. Therefore, in Chapter 2.4, an auxiliary pressure vessel was proposed to suppress bubble generation by maintaining the pressure at the inlet of the PR pump without reducing the number of PR bottles or placing them upward. This was applied to a newly designed facility along with software featuring a dedicated control algorithm to establish a new PR dispensing system.

The auxiliary pressure vessel mentioned above can effectively reduce the pressure change of PR moving in the tube through the use of a control algorithm that pressurizes and decompresses the chemical solution through the contraction and expansion of the inner tube. This helps to effectively reduce the generation of bubbles. The results of the pressure profile check for the pump refill section after the chemical liquid discharge, without using the proposed auxiliary pressurization device in this study. The pressure at the pump inlet gradually decreases to approximately -5 kPa to transfer the chemical liquid from the PR bottle to the internal stage of the pump using differential pressure. This pressure change creates the worst conditions for bubble formation in the entire pump operation sequence.

On the other hand, the pressure profile obtained by applying the proposed auxiliary pressurization device with an effective algorithm to correct PR pressure during the reload operation of the pump. It can be observed that the PR reload operation is carried out while maintaining a stable pressure of about 15 kPa at the pump inlet with minimal pressure changes. This not only suppresses the generation of bubbles by preventing the negative pressure of PR and maintaining a positive pressure, but also prevents the generation of additional bubbles by suppressing effective pressure fluctuations in the overall operating cycle condition.

In conclusion, by analyzing the root causes and implementing effective improvements, restrictions on the installation location and quantity of the PR bottles can be removed, providing design flexibility to load a large amount of the PR bottles in various locations.

4 Conclusion

In the semiconductor lithography process, chips can be manufactured with a narrow line width as the technology of a scanner facility that performed exposure develops, and for this reason, management of nano particles is of utmost importance. This study was conducted to improve the management of smaller wet particles and air bubbles in the photo track facility with these process technology advances. The root cause of air bubbles, a chronic problem in the photo process, was found to be the lack of pressure management. The fluid pressure in the PR dispensing system had to be managed with a more positive pressure, and the pressure change had to be further reduced during the transportation of the chemical liquid.

Overall, based on these improved design methods, a method of improving existing equipment that utilizes the potential energy of the PR bottles was devised and implemented. Additionally, new parts were invented and proposed that can increase the pressure at weak points, along with control algorithms, and were applied to the new facility. Through these comprehensive activities, we have continuously improved product quality and productivity. We hope that the results of this study, which identified quantitative trends based on a physical understanding of the facility, will serve as a valuable reference for managing critical particles and bubbles at the 15–17 nm level in the future.

References

1. Fissan, H., et al.: Basic Solutions for Nanoparticle Contamination in EUV-Lithography. *VDI Berichte* **1920**, 11–14 (2005)
2. Kamei, Y., et al.: Technology for defectivity improvement in resist coating and developing process in EUV lithography process. *Proc. SPIE* 10143–73 (2017)
3. Wagener, C., Harned, N.: Lithography gets Extreme. *Nat. Photonics* **4**, 24–26 (2010)
4. Brakensiek, N., Sevegney, M.: Effects of dispense equipment sequence on process start-up defects. *Proc. SPIE* 8682 (2018)
5. Couteau, T., et al.: Lithography Cost Savings Through Resist Reduction and Monitoring Program. *IEEE/SEMI ASMC* (2011)
6. Li, B., et al.: “A sandwiched flexible polymer mold for control of particle-induced defects in nanoimprint lithography”, *Applied physics. A, Materials science & processing* **110**, 123–128 (2013)
7. Chunmeng, L., et al.: Research on heat dissipation of car seat optimizing based on ARIZ analysis. *Journal of Physics, Conf. Series* 1303 (2019)